THE INTEGRAL CECH COMPLEX TENSORED WITH ANY PRESHEAF OF RINGS STAYS EXACT IN POSITIVE DEGREES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed. The author offers a second proof route with the words 'Or you can use'; both routes are kept.

Editorial note. The counted statement opens with the author's own slogan environment. The chapter's retained minimal preamble does not carry the project's no-op definitions of the display environments, so the documented no-op definitions of reference, slogan and history are added as a separate verbatim preamble span taken from the snapshot (sources/stacks-varieties/retained-preamble.tex lines 75-77, themselves the project's own wrapper definitions). Nothing was invented and no macro was redefined.

Editorial note. Same-chapter references are located at their \label lines inside the pinned chapter file; the cross-chapter pointer is located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim line carrying its canonical tag.

Allowed standard prerequisites (editorial).

- The Cech chain complex of a family and the complex of modules it represents. Same chapter, Lemma labelled lemma-cech-map-into (source line 971, Stacks tag 03AS, counted earlier in this night's work); named in the counted statement at line 1167. Supplies the complex Z_{U, bullet} being tensored; the exactness statement about it is the counted content.
- Structure of the integral Cech complex at an object: a direct sum of complexes homotopic to Z in degree zero. Same chapter, Lemma labelled lemma-homology-complex (source line 1025, Stacks tag 03AT, already counted by the card DSH-alg-cech-homology-exact); invoked at proof lines 1174 and 1184. Supplies the description used in both branches of the proof; the tensor computation on top of it is retained in the counted span.

Source reference. Original lemma 03AS: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 971–971. Same-chapter pointer named in the counted statement at line 1167 as the definition of the complex being tensored; the exactness argument is retained verbatim in the counted span.

Source reference. Original lemma 03AT: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1025–1025. Same-chapter pointer invoked at proof lines 1174 and 1184 for the shape of the integral Cech complex at an object; the tensor and exactness arguments are retained in the counted span.

Source reference. Original lemma 03EZ: sources/stacks-chow/tags.txt, source lines 4474–4474. Cross-chapter standard prerequisite offered at proof lines 1181-1182 as the alternative route through flatness of the terms and of H_0; the first route through homotopy is retained in the counted span.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *The integral presheaf Čech complex is a flat resolution of the constant presheaf of integers. Let $\mathcal{C}$ be a category. Let $\mathcal{U} = \{f_i : U_i \to U\}_{i \in I}$ be a family of morphisms with fixed target such that all fibre products $U_{i_0} \times_U \ldots \times_U U_{i_p}$ exist in $\mathcal{C}$. Let $\mathcal{O}$ be a presheaf of rings on $\mathcal{C}$. The chain complex*

$$\mathbf{Z}_{\mathcal{U},\bullet} \otimes_{p,\mathbf{Z}} \mathcal{O}$$

is exact in positive degrees. Here $\mathbf{Z}_{\mathcal{U},\bullet}$ is the chain complex of Lemma 03AS, and the tensor product is over the constant presheaf of rings with value $\mathbf{Z}$.

Proof. Let V be an object of $\mathcal{C}$. In the proof of Lemma 03AT we saw that $\mathbf{Z}_{\mathcal{U},\bullet}(V)$ is isomorphic as a complex to a direct sum of complexes which are homotopic to $\mathbf{Z}$ placed in degree zero. Hence also $\mathbf{Z}_{\mathcal{U},\bullet}(V) \otimes_{\mathbf{Z}} \mathcal{O}(V)$ is isomorphic as a complex to a direct sum of complexes which are homotopic to $\mathcal{O}(V)$ placed in degree zero. Or you can use Modules on Sites, Lemma 03EZ, which applies since the presheaves $\mathbf{Z}_{\mathcal{U},i}$ are flat, and the proof of Lemma 03AT shows that $H_0(\mathbf{Z}_{\mathcal{U},\bullet})$ is a flat presheaf also. □